

Solution

FORCE AND PRESSURE -03

Class 08 - Science

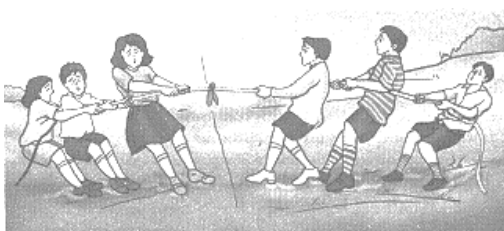
1. The force acting on a unit area of a surface is called pressure. To find pressure, therefore, we need to divide force by the surface area on which it acts.
So, pressure = $\frac{\text{Force}}{\text{Area}}$
2. Electrostatic force is non-contact force.
3. The two forces acting on rocket are :
 - (i) Force due to gravity acts vertically downward.
 - (ii) Force of push acting upward.
4. The force by which earth or any other object attracts objects towards itself is called the gravitational force.
5.
 - (i) Electrostatic forces
 - (ii) Magnetic force
 - (iii) Force due to gravity.
6. The two categories of forces are:
 - (i) contact forces
 - (ii) non-contact forces.
7. (d) atmospheric pressure.
8. Liquids also exert pressure. When a liquid is put in a vessel it exerts pressure on its bottom and wall.
9. The envelop of air around us is called atmosphere.
10. To increase pressure we should exert the same force on a smaller area.
11. There is no air inside the two hemispheres, so only outer surface is in contact of atmospheric pressure. This is because we cannot separate them.
12. The force acting on a unit area of a surface is called pressure.
Pressure = $\frac{\text{Force}}{\text{Area}}$
13. Forces applied on an object in the same direction add to one another.
14. The pressure exerted by the air is known as atmospheric pressure.
15. The gravitational force is non-contact force.
16. A push or a pull on an object is called force.
17. Due to gravitational force.
18. When an inflated balloon rubbed with a piece of synthetic cloth, it becomes charged. A charged body attracts an uncharged body.
When this charged balloon is pressed against a wall, it sticks to the wall. Thus, the electrostatic force acts between the charged balloon and the wall.
19. If the two forces act in the opposite direction on an object, the net force acting on it is the difference between the two forces.
20. Yes, liquids and gases also exert pressure.
21. It's speed decrease because the force due to gravity opposes its upward motion.
22. Muscular force is exerted to change the shape of the dough.
23. The envelop of air around the earth is known as atmosphere. The atmospheric air extends up to many kilometres above the surface of the earth. The pressure exerted by the air is called atmospheric pressure.
24. Here, the suspended load exerts a force on the spring and pushes the spring downwards. As a result, the spring gets stretched.
Hence, its shape gets changed.
25. The force brings about a change in the shape of hot iron piece.
26. Various big or small actions make us feel the force. We hit or catch many objects in our daily life. We see that a moving ball stops on its own. The ball changes the direction of its motion when it is hit with a bat. We make lassi etc. by churning of curd. These are many actions which help us to feel that a force is exerted.
27. Force has both magnitude and direction.
28. No, gravity is not a property of earth only.
29. Frictional force is responsible for wearing out of bicycle tyres
30. The feet of the athlete exerts the muscular force on the ground, which pushes the ground. This allows him to jump over the bar. As a result, his state of motion gets changed.
31. We make a muscular force to extract the juice of the lemon by squeezing it. As a result, the shape of the lemon gets changed.

32. A liquid exerts pressure on the walls of the container.
33. Yes, gases also exert pressure on the walls of the containers.
34. To take out paste from the toothpaste tube we use our muscular force. This muscular force acting on the toothpaste tube leads to a change of its shape.
35. Magnetic force act on to sort out pins and, it is a non-contact force.
36. The upward force is larger than the force of gravity because the balloon is moving upward.
37. When a force is applied on an object it can.
 - (i) Change the shape of the object.
 - (ii) Change in the state of motion of the object.
38. Take an empty plastic bottle. Drill four holes all around near the bottom of the bottle. Make sure that all the holes are at the same height from the bottom. Now fill the bottle with water. We observe that different streams of water coming out of the holes fall at the same distance from the bottle. This observation indicates that liquids exert equal pressure at the same depth.



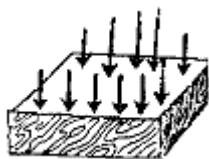
Liquid exerts equal pressure at the same depth.

39. The electrostatic force is responsible for this phenomenon. As we know that electrostatic force is the force exerted by a charged body on another charged or uncharged body. Hence, during dry weather, clothes made of synthetic fiber (uncharged) often stick to the skin (charged).
40. Her friend should push the box to the east so that it moves faster in the same direction because:
 - i. If her friend pushes the box in the east direction, the magnitude of the force increases.
 - ii. As a result, the box will move faster.
41. Muscular force: The force resulting due to the action of muscles is known as muscular force. Since muscular force can be applied only when it is in contact with an object, it is also called a contact force.
42. We know that if two forces act on opposite direction on an object, the net force acting on it is the difference between the two forces.
If the two teams pull the rope with equal force, the forces will be a opposite directions and will cancel each other and the rope will not move.



43. Yes, the force of gravity acts on dust particles.
 - i. Force of gravity is an attractive force.
 - ii. It acts on all objects.
 - iii. The force of gravity acts on all of us all the time without being aware of it.
44. We know that air is present everywhere. The pressure of air inside our body is same as that of the atmosphere. Therefore, we do not experience its effect. Isn't nature kind to us!
45. When a force is applied on an object, it may change its speed. If the applied force is in the direction of motion, the speed of the object increases. If the force is applied in the direction opposite to the motion, then it results in a decrease in the speed of the object.
46. (i) Making a chapati from a ball of dough.
(ii) Hanging a weight with a rubber band.
(iii) By pressing the clay between the hands, it deforms.
(iv) The shape of the plastic bottle changes by squeezing it. (any two)

47. When one surface is moving over another surface, a force comes into play and opposes their relative motion. This force is called friction or force of friction. Friction always opposes motion because it acts in opposite direction to motion. Force of friction arises due to contact between two surfaces. So friction force is called a contact force.
48. The effect of a force depends on two things.
- The amount of force applied. The more the force applied the greater the effect.
 - The area on which the force is applied. The smaller the area the force is applied on, the greater the effect.



49. There will be two forces acting on the rocket:
- Upward force applied by the rocket engine.
 - Downward gravitational force applied by the earth.
50. Gravitational force pulls them down. Objects or things fall towards the earth because it pulls them. This force is called the gravitational force. Hence, while sieving grains, small pieces fall down due to the gravitational force of the earth.
51. Force is a pull or push of the objects. Force can make a stationary object move or make a moving object move faster. The action like, pushing, pulling, picking, hitting, lifting, running and bending are the examples of force. Moving or stopping of a body, changing shape and direction of motion of objects are various actions which show force in play.
52. i. When two forces are exerted in same direction on an object, then the forces are added and action becomes easy.
 ii. When two forces act in the opposite directions on an object, the net force acting on it is the difference between the two forces.



(a)



(b)

Two friends pushing a heavy load (a) in the same direction, (b) in opposite directions.

53. When water is filled in the bucket, the water exerts pressure at the bottom of the bucket the most. As we know that pressure in a liquid increases with the depth, so the pressure would be more at the bottom. Hence water comes out from the bottom much faster than the upper hole.
54. **Push:**
- To close a drawer
 - To move a wooden box on floor
- Pull:**
- To open a drawer
 - A horse pulling a cart

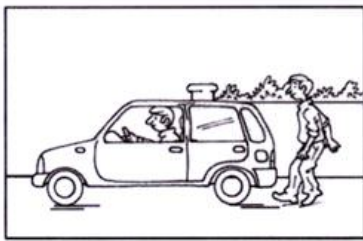
55. Pressure is defined as the force acting per unit area.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

Note that the pressure is inversely proportional to the area. So smaller the area, larger the pressure on a surface.

Chapatti maker works on this principal. This large pressure converts balls of dough into chapattis

56. D, B, A, C, because pressure of a liquid column depends upon the height of the liquid column and not on volume of the liquid.
57. In this case, muscular force is acting in upward direction and the force of gravity is acting in downward direction. Both forces are acting in opposite directions to each other and hence nullify the effect of each other. Due to this, there is no change in the state of motion of the bucket.
58. The state of motion of an object is described by its speed and the direction of motion. The state of rest is considered to be the state of zero speed. An object may be at rest or in motion, both are its state of motion. Any change in the speed, or direction of motion, or both, implies a change in the state of motion of the object.
59. Yes, the force of gravitation exists between two astronauts in space because:
- Every object in the universe whether small or large exerts a force on every other object.
 - We can say that this is the universal law of gravitation.
60. Imagine, a man is standing behind a stationary car. The car does not move due to his presence.



(a) A man standing behind a stationary car.



(b) A car being pushed by a man.

Now allow the man to push the car, it means he applies a force on the car. The car may begin to move in the direction of the applied force. Note that the man to push car to make it move. This example shows that at least two objects must interact with each other for a force to come into play.

61. Pressure is produced when a force acts on an object. It is defined as the force acting normally on a unit area of an object.

The mathematical expression for force is:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

The SI unit of pressure is newton per square metre which is also called pascals(Pa).

62. a. The layer of air above the earth is called atmosphere. The air has weight, so it exerts a pressure on the surface of the earth, which is called atmospheric pressure.
Atmospheric pressure is due to the weight of air present in the atmosphere above us. It acts in all directions.
- b. Our body has blood which flows in every cell through blood vessels. Our blood itself exerts a pressure that is greater than the atmospheric pressure. Since atmospheric pressure from outside our body is balanced by the blood pressure acting from inside, we don't get crushed.
63. A dropper is a short glass tube with a rubber bulb at one end and a nozzle at the other end. It is used for measuring the drops of liquids as in the case of medicines. It works on the principle of atmospheric pressure.
When we press the rubber bulb, the air present in the glass tube escapes from the nozzle in the form of bubbles. The air pressure inside the rubber bulb and the nozzle is reduced and a greater atmospheric pressure is acting on the surface of the liquid which pushes the liquid inside the glass tube. This is how it works on the principle of the atmospheric pressure.
64. The woman wearing sandals with flat soles will feel more comfortable while walking on the sandy beach because the sandals with flat soles have a larger area as compared to the pointed heels. As we know that pressure is inversely proportional to the area. Hence, the larger the area, the smaller will be the pressure. Therefore, the pressure exerted by the sandals having flat soles will be less. On the other hand, the pressure exerted by the pointed heels will be more. That's why walking with flat soles will be more comfortable.
65. The drinking straw is a thin pipe used for drinking cold drinks. It works on the principle of the atmospheric pressure. The lower end of the straw is inside the liquid, so when we suck at the upper end with our mouth, the pressure inside the straw and in our mouth is decreased. But the pressure exerted at the surface is greater which pushes the liquid up and it comes inside our mouth.
66. a. A school bag should have wider straps so that the weight of the bag fall over a large area of the shoulder of the child which will produce less pressure on the shoulder of the child. Hence it would be comfortable for the child to carry a heavy bag.
b. A sharp knife has a very thin edge to its blade. It cuts better because due to its very thin edge, the force exerted by our hands will fall over a very small area of the object producing a large pressure that cuts the object easily.
67. The various effects of force are as follows:
- A force can change the state of motion, i.e. it can move a stationary body and it can stop a moving body also.
 - A force can change the speed of an object.
 - A force can change the direction of a moving body.
 - A force can change the shape and size of a body.
68. The atmospheric pressure above the surface of the earth is maximum at the sea level. It is because air column above us is the highest at the sea level. As we go up in the atmosphere from the surface of the earth, the atmospheric pressure goes on decreasing. It is because the weight of air goes on decreasing due to which the pressure also decreases. Hence the pressure at the top of the mountain will be much less than at the base.
69. It is easier to walk with flat shoes on the soft sand because the flat shoes have a greater area in contact with the sand due to which there will be less pressure on the soft sand. On the other hand shoes with small heels exert large pressure on the sand because of its small area, hence it would be difficult for us to walk on the sand as the heels will sink into the soft sand.
70. The initial force that sets the arrow in motion is a muscular force:
- This force is caused by the action of muscles in our body.
 - The archer stretches his muscles to stretch the string of the bow.

- iii. When he releases strings, the shape of the bow changed.
- iv. As soon as the string released, the string regains its original position.
- v. It provides the initial force to set the arrow in motion.

The arrow ultimately falls down due to force of gravity:

- i. The force of gravity always acts in a downward motion.
 - ii. The arrow ultimately falls towards the earth because the earth pulls it.
 - iii. Force of gravity is an attractive force.
71. When a man stands on a cushion then his two feet are in contact with the cushion. Due to this his body weight falls on a small area. On the other hand when a man lies on the cushion, his whole body comes in contact with the cushion. His weight falls over the large area because of which the depression in the cushion is less than when he stands on it.
72. a. The pressure of the liquids is not same at all depths. It changes with the depth.
The pressure exerted by a liquid increases with increasing depth inside the liquid. As the depth increases, the weight of liquid column pushing down from above increases and hence the pressure increases too.
- b. The walls of a dam are thicker at the bottom because the pressure exerted at the bottom is more than the top, so to bear the pressure and the weight of the liquid, the walls are made thicker.
73. When a bucket full of water is held in our hand above the ground, there will be a muscular force applied by the hand and downward gravitational force applied by the earth. There will be no change in the state of motion because the two forces are equal and opposite to each other, hence they will balance each other.
74. a. A force which can be exerted by an object on another object only through some contact is called a contact force. The examples of contact forces are: muscular force and frictional force.
Since muscular force is applied on an object with our muscles, hence it is a contact force. For example, a boy pulling a cart. Frictional force is also applied between the two surfaces in contact with each other, for example, a car moving on the road is stopped by applying friction.
- b. A force which can be exerted by an object on another object even from a distance without touching each other is called a non contact force.
Magnetic force and gravitational force are examples of non contact forces. A magnet can attract an iron object even at a distance, so it is a non contact force.
Gravitational force is also applied on all objects even at a distance, so it is a non contact force. For example, an apple falls from a tree comes to the earth in the downward direction.
75. The force exerted by the electrically charged object is called electrostatic force.
The electrostatic force can be exerted by a charged body on another body even from a distance when they are not in contact with each other, hence it is also called a non contact force.
For example, when a plastic comb is rubbed on dry hair and brought near the tiny pieces of paper, we will see that the tiny papers are attracted towards the comb. It is due to the electrostatic force exerted by the electric charges acquired by the comb.